Methodology

Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Optimization in Smart Cities

21.02.2026

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1. Overview

This research adopts a **simulation-driven experimental methodology** to design, implement, and evaluate an adaptive traffic signal control system based on **Multi-Agent Reinforcement Learning (MARL)**.

The methodology is structured into three core components:

1. Simulation Environment Design
2. Baseline System Implementation
3. MARL-Based Adaptive System (in progress)

A comparative evaluation framework is used to assess system performance under controlled traffic conditions.

2. Simulation Environment Design

2.1 Environment Structure

The traffic system is modeled as a **grid-based network of intersections**, where:

- Each node represents a traffic signal (intersection)
- Each edge represents a road segment
- Vehicles move along predefined paths

Initial configuration:

- Grid size: 4 × 4 intersections
- Four-directional traffic flow (N, S, E, W)
- Discrete-time simulation steps

2.2 Traffic Modeling

Traffic is modeled using stochastic generation patterns to simulate real-world variability.

Vehicle Generation:

- Randomized arrival rates
- Configurable traffic density (low, medium, high)

Movement Rules:

- Vehicles follow lane discipline
- Movement constrained by signal states
- Queue formation at intersections

2.3 State Representation (Environment Level)

At each timestep, the environment captures:

- Queue length per lane
- Waiting time per vehicle
- Signal phase status
- Traffic density per intersection

This information is used by both baseline and learning agents.

3. Baseline System (Implemented)

The baseline system serves as a **control mechanism for comparison**.

3.1 Fixed-Time Signal Control

- Predefined signal phases
- Fixed duration for each phase
- No adaptation to real-time traffic

3.2 Signal Cycle Example

- North-South Green → Fixed duration
- East-West Green → Fixed duration
- Transition phases (yellow/red buffer)

3.3 Purpose of Baseline

- Establish performance benchmark
- Provide comparative metrics
- Validate simulation correctness

4. MARL-Based System (In Progress)

The proposed system models each intersection as an independent **reinforcement learning agent**.

4.1 Agent Definition

Each intersection operates as:

- An autonomous decision-making unit
- Observing local traffic conditions
- Selecting optimal signal actions

4.2 State Space

Each agent observes a localized state:

- Queue length per incoming lane
- Average waiting time
- Current signal phase
- Traffic density indicators

State representation may be encoded as:

- Numerical vectors
- Normalized values for stability

4.3 Action Space

At each decision step, the agent can:

- Select next signal phase
- Optionally adjust phase duration

Example actions:

- Switch to North-South green
- Switch to East-West green

- Maintain current phase

4.4 Reward Function

The reward function is designed to optimize traffic efficiency:

Primary Objectives:

- Minimize total waiting time
- Minimize queue length
- Maximize vehicle throughput

Example Reward Formulation:

- Negative reward for long queues
- Positive reward for clearing vehicles

4.5 Learning Strategy

The MARL system will follow a **decentralized learning approach**:

- Each agent learns independently
- No explicit inter-agent communication (initial phase)

Potential algorithms:

- Q-Learning (baseline RL)
- Deep Q Networks (DQN)
- Policy Gradient methods (future extension)

4.6 Training Process

- Simulation episodes represent traffic scenarios
- Agents interact with the environment
- Rewards guide policy updates

Training considerations:

- Exploration vs exploitation balance
- Convergence monitoring

- Stability of learning

5. Evaluation Framework

The system will be evaluated using consistent experimental conditions.

5.1 Metrics

Performance will be measured using:

- Average waiting time per vehicle
- Average queue length
- Throughput (vehicles processed per timestep)
- Congestion index

5.2 Experimental Scenarios

The following scenarios will be tested:

- Low traffic density
- Medium traffic density
- High traffic density
- Dynamic traffic patterns

5.3 Comparative Analysis

Two systems will be compared:

1. Fixed-Time Baseline System
2. MARL-Based Adaptive System

Evaluation will focus on:

- Performance improvement
- Stability under varying conditions
- Generalization across scenarios

6. Scalability Evaluation Strategy

Rather than large-scale simulation initially, the study adopts a **controlled scalability approach**:

- Start with 4×4 grid
- Extend to 6×6 or 8×8 grid
- Observe performance degradation or stability

This allows analysis of:

- Agent coordination under increased complexity
- System robustness

7. Implementation Stack

The system is implemented using:

- Simulation: Custom grid-based simulator
- Backend: Python
- Visualization: Real-time dashboard
- Data handling: Structured logs for analysis

Future extensions may include:

- Integration with simulation tools (e.g., SUMO)
- GPU-based training acceleration

8. Assumptions and Constraints

Assumptions:

- Vehicles follow simplified movement rules
- No accidents or unexpected disruptions
- Homogeneous vehicle behavior

Constraints:

- Limited real-world data integration (current phase)

- Computational limits for large-scale simulations

9. Summary

This methodology establishes a **controlled, reproducible experimental framework** to evaluate adaptive traffic control using MARL.

By starting with a baseline system and incrementally introducing learning-based agents, the study ensures:

- Clear performance comparison
- Structured experimentation
- Scalable research progression

The approach balances **practical implementation** with **research rigor**, enabling future expansion into real-world applications.